IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

SUSTAINABLE ELECTRICAL SYSTEMS

Thursday 11 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper. Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : G. Strbac, B. Chaudhuri
Second Marker(s) : B. Chaudhuri, G. Strbac

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Answer ALL Questions

1. Question on Wind and Photovoltaic Power

a) A 9.5 MW direct-drive wind turbine with a full rated converter and permanent magnet synchronous generator (PMSG) has 85 m long blades. The maximum turbine efficiency is 47% at an optimum tip-speed ratio of 8.5. The cut-in and cut-out speeds of the wind turbine are 3 m/s and 22 m/s, respectively. Considering the air density to be 1.22 kg/m^3 and neglecting losses in the generator, calculate the following:

i) rated wind speed (in m/s).

[1]

ii) maximum power (in MW) that can be generated by the wind turbine at wind speeds of 6 m/s and 12 m/s.

[2]

iii) range of rotational speed (in rpm) of the wind turbine required to extract maximum power from wind.

[3]

iv) the rotational speed of the turbine (in rpm) for a wind speed of 12 m/s.

[2]

v) minimum number of poles needed for the PMSG if the minimum allowed stator frequency is 8 Hz.

[2]

b) A developer is setting up a solar PV array at a location having a latitude of 40°N. The plan is to have 500 solar PV modules each with a rated power of 330 W and 18% efficiency under standard test conditions (STC). A PV array is formed with 10 parallel strings with each string having 50 PV modules in series with a string diode. Assume all PV modules to be operating at their maximum power point for the available irradiance.

i) What is the approximate surface area (in m^2) of each PV module?

[2]

ii) What is the total power output (in kW) of the PV array when the irradiance is 600 W/m^2 assuming the same efficiency as under STC?

[1]

iii) How much power (in kW) is generated by the PV array under STC if 7 PV modules in 3 separate strings are completely shaded?

[2]

iv) How much power (in kW) is generated by the PV array under STC if 2 PV modules within 1 particular string are completely shaded?

[1]

v) What should the tilt angle (in degrees) generally be, if the PV modules were to have a fixed tilt and what factors should be considered while deciding it?

[2]

vi) If the PV modules were to have a single-axis north-south tracker, what should be the tilt angle (in degrees) of the PV modules on the summer and winter solstice days?

[2]

2. Question on balancing, Voltage Control and Security of Supply

Part A

a) Explain the techniques for providing reserve.

[1]

b) Explain the advantages and disadvantages of using flexibility to offset the need for reserve from conventional generation.

[3]

Part B

a) Explain passive approach for distributed generation connection and active approaches for increasing network distributed generation hosting capacity?

[2]

b) What is the maximum active power of distributed generation with unity power factor that could be connected to the end of a 11 kV circuit when the reactive compensation connected at generation busbar is 1 MVAr? Circuit resistance is 0.65 Ohms and reactance is 0.437 Ohms. Voltage at beginning of the circuit is 11 kV and maximum voltage deviation in 11 kV network is 6%. Answer should contain a sketch of the considered system with denoted relevant variables.

[2]

(Question 2 continues on the next page)

Part C

Contribution of distributed generations to the security of supply depends on the demand profiles and availability of the resources. In this case, three units, connected to busbars A can secure proportion of demand supplied from busbar A as shown in Figure 2.1. Those units have rating of 10, 5 and 2 MW and availability of 85, 75 and 65%, respectively.

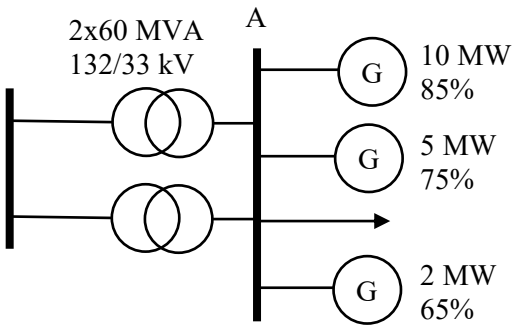


Figure 2.1 Diagram of the considered part of the network

Demand profile during a peak season, that lasts 100 days, is represented by the normalised load duration curve as shown in Figure 2.2.

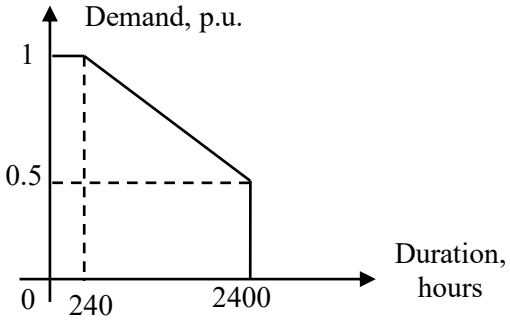


Figure 2.2 Normalised load duration curves for two characteristic days

a) Form the capacity outage probability table for plants providing network support at busbars A by stating capacity in service and the corresponding probability value for each state.

[4]

b) Calculate the Expected Energy Not Supplied of the system supplied by the generation plant using above load duration curve.

[5]

c) Calculate the capacity of the equivalent source, i.e. firm capacity, that never fails that results in the same Expected Energy Not Supplied.

[3]

3. Question on Low Carbon Heat

a) In 2008 the UK Government passed legislation entitled the Climate Change Act 2008. This required a 80% reduction in the UK's greenhouse gas emissions (GHG) by 2050 compared to emissions in 1990. The Government also established the Committee on Climate Change (CCC).

i) Briefly summarise the purpose of the CCC.

[2]

ii) In June 2022 the CCC presented its report to Parliament "Progress in reducing emissions". Outline the reason for the report and the key messages.

[2]

b) In 2019 the UK introduced legislation to bring all greenhouse gas emissions to net zero by 2050. This compares to its previous targets of at least a 80% reduction from 1990 levels. In order to meet this target it is likely that removal of CO_2 from the atmosphere will be required. Two technologies for removing CO_2 from the atmosphere are:

- Bioenergy with carbon capture and storage (BECCS)
- Direct air capture of CO_2 with storage (DACCS)

i) Briefly explain why CO_2 removal is likely to be required.

[1]

ii) Give a description of BECCS and DACCS.

[2]

c) Give a brief description of the following types of low carbon heating systems.

i) Air source heat pump

[2]

ii) Hybrid heat pump

[1]

and

iii) Outline one benefit and one drawback of a hybrid heat pump over a ASHP?

[2]

iv) Explain how a heat pump achieves an efficiency exceeding 100%.

[1]

d) List the factors that determine building space heat demand.

[1]

e) A building has a floor area of 120 m^2 and an average thermal transmittance of 2.0Wth/m2K. In a typical year the building's space heating is on for 100 days, 10 hours each day. The internal temperature is 18°C and the average external temperature when the space heating is on 5.5°C.

i) Calculate the annual space heating demand in a typical year in MWh. State any assumptions made.

[3]

ii) Calculate the peak heat demand using the same assumptions but when the external temperature is -7°C.

[2]

iii) If the building energy efficiency is improved by 20% what is the reduction in peak heat demand?

[1]

4. Question on Flexibility, Hydro Power and Offshore Transmission

Demand Flexibility

a) Explain the phenomenon of demand payback associated with using flexible demand in DSR applications. Discuss how demand payback affects the volume and scheduling of DSR flexibility as opposed to other forms of flexibility such as energy storage.

[2]

b) Heat demand diagram for a household during 10 contiguous half-hourly intervals on a cold winter evening is plotted in Figure 4.1. The household's heat demand is delivered by an air-source heat pump (ASHP) system with a Coefficient of Performance (COP) equal to 2.0. Answer the following questions:

i) What is the peak electricity demand for the household if we ignore any non-heating electricity consumption? Plot the household's electricity demand diagram during the 5-hour period (i.e. over the 10 half-hourly intervals) shown in Figure 4.1.

[2]

ii) Calculate the size of heat storage (in kWh_{th}) that would be necessary to limit the household's peak electricity demand for heating to 5 kW_{el}, if the energy-to-power ratio of the heat storage (i.e. its energy capacity in kWh_{th} divided by its heat charge/discharge rate in kW_{th}) is fixed and equal to 2. Plot the household's new electricity demand diagram during the period shown in Figure 4.1.
(Assume that the heat storage is fully charged by 16:00 and can be fully discharged by 21:00. Ignore any heat losses associated with storing and releasing heat from thermal storage.)

[4]

iii) Repeat the calculation of the heat storage size from question ii) (using the same assumptions) if the household installs a more efficient heat pump with a COP of 2.5. (It is not necessary to plot the new electricity demand diagram.)

[2]

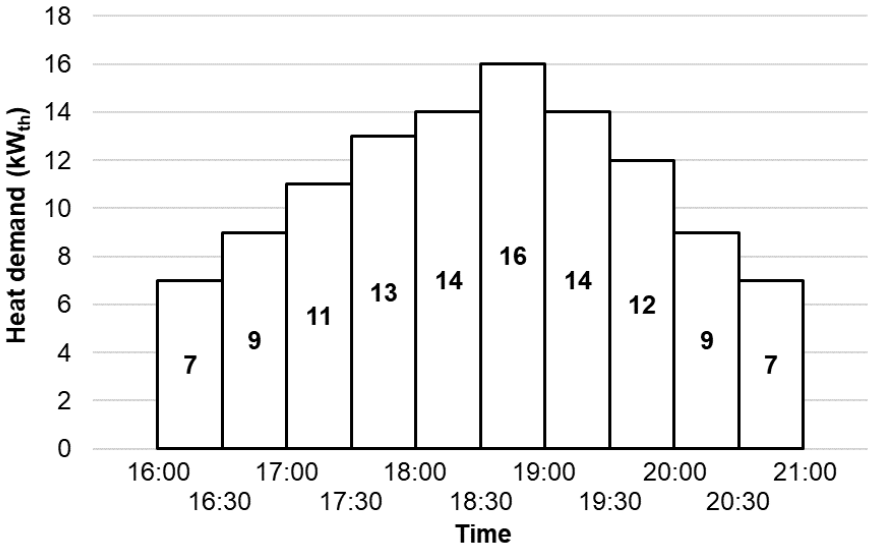


Figure 4.1. Household heat demand during peak demand period

Hydro Power

c) A reservoir-based hydropower plant has its headwater and tailwater levels at 120 m and 30 m, respectively. Each generator in the hydropower plant is rated at 600 MW operating at 50 Hz with a rated speed of 100 rpm. There are 15 such generators. The efficiency of the plant is 95%.

i) What is the effective head (in m)?

[1]

ii) How much volume flow rate of water (in m^3/s) is needed to produce the rated power of the hydropower plant?

[2]

iii) How many poles are needed for each generator?

[2]

Offshore Transmission

d) Explain why HVDC is preferred over traditional AC for bulk power transmission for long distances through sub-sea cables.

[3]

e) Explain why polymeric cables which are stronger and lighter can only be used for VSC HVDC but not LCC HVDC.

[2]